

Languages

Language: a set of strings

String: a sequence of symbols
from some alphabet

Example:

Strings: cat, dog, house

Language: {cat, dog, house}

Alphabet: $\Sigma = \{a, b, c, \dots, z\}$

Languages are used to describe
computation problems:

$$PRIMES = \{2, 3, 5, 7, 11, 13, 17, \dots\}$$

$$EVEN = \{0, 2, 4, 6, \dots\}$$

Alphabet: $\Sigma = \{0, 1, 2, \dots, 9\}$

Alphabets and Strings

An alphabet is a set of symbols

Example Alphabet: $\Sigma = \{a, b\}$

A string is a sequence of symbols from the alphabet

Example Strings

a

ab

abba

aaabbbbaaba b

u = ab

v = bbbbaaa

w = abba

Decimal numbers alphabet $\Sigma = \{0,1,2,\dots,9\}$

102345

567463386

Binary numbers alphabet $\Sigma = \{0,1\}$

100010001

101101111

Unary numbers alphabet $\Sigma = \{1\}$

Unary number: 1 11 111 1111 11111

Decimal number: 1 2 3 4 5

String Operations

$$w = a_1 a_2 \cdots a_n$$

abba

$$v = b_1 b_2 \cdots b_m$$

bbbbaaa

Concatenation

$$wv = a_1 a_2 \cdots a_n b_1 b_2 \cdots b_m$$

abbabbbbaaa

$$w = a_1 a_2 \cdots a_n$$

ababaaaabbb

Reverse

$$w^R = a_n \cdots a_2 a_1$$

bbbbaaababa

String Length

$$w = a_1 a_2 \cdots a_n$$

Length: $|w| = n$

Examples: $|abba| = 4$

$$|aa| = 2$$

$$|a| = 1$$

Length of Concatenation

$$|uv| = |u| + |v|$$

Example: $u = aab, \quad |u| = 3$

$v = abaab, \quad |v| = 5$

$$|uv| = |aababaab| = 8$$

$$|uv| = |u| + |v| = 3 + 5 = 8$$

Empty String

A string with no letters is denoted: λ or ε

Observations: $|\lambda| = 0$

$$\lambda w = w\lambda = w$$

$$\lambda abba = abba\lambda = ab\lambda ba = abba$$

Substring

Substring of string:

a subsequence of consecutive characters

String

abbab

abbab

abbab

abbab

Substring

ab

abba

b

bbab

Prefix and Suffix

abbab

Prefixes

Suffixes

λ

abbab

a

bbab

ab

bab

abb

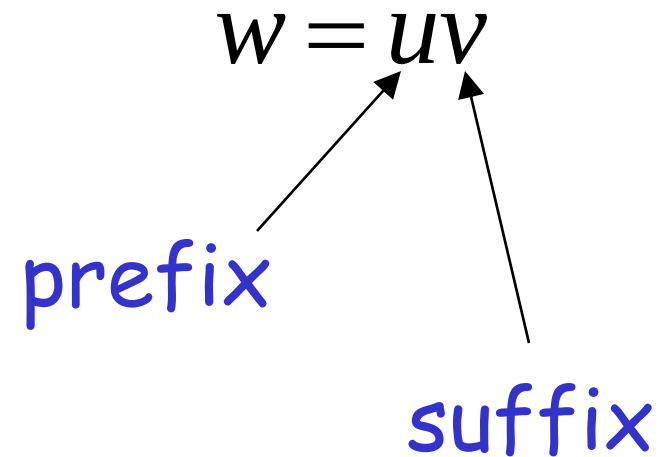
ab

abba

b

abbab

λ



Another Operation

$$w^n = \underbrace{ww \cdots w}_n$$

Example: $(abba)^2 = abbaabba$

Definition: $w^0 = \lambda$

$$(abba)^0 = \lambda$$

The * Operation

Σ^* : the set of all possible strings from
alphabet Σ

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \dots\}$$

The + Operation

Σ^+ : the set of all possible strings from alphabet Σ except λ

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \dots\}$$

$$\Sigma^+ = \Sigma^* - \lambda$$

$$\Sigma^+ = \{a, b, aa, ab, ba, bb, aaa, aab, \dots\}$$

Languages

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, \dots\}$$

Language: $\{\lambda\}$

Language: $\{a, aa, aab\}$

Language: $\{\lambda, abba, baba, aa, ab, aaaaaa\}$

More Language Examples

Alphabet $\Sigma = \{a, b\}$

An infinite language $L = \{a^n b^n : n \geq 0\}$

λ
 ab
 $aabb$
 $aaaaabbbbb$

} $\in L$ $abb \notin L$

Prime numbers

Alphabet $\Sigma = \{0,1,2,\dots,9\}$

Language:

$PRIMES = \{x : x \in \Sigma^* \text{ and } x \text{ is prime}\}$

$PRIMES = \{2,3,5,7,11,13,17,\dots\}$

Even and odd numbers

Alphabet $\Sigma = \{0,1,2,\dots,9\}$

$EVEN = \{x : x \in \Sigma^* \text{ and } x \text{ is even}\}$

$EVEN = \{0,2,4,6,\dots\}$

$ODD = \{x : x \in \Sigma^* \text{ and } x \text{ is odd}\}$

$ODD = \{1,3,5,7,\dots\}$

Unary Addition

Alphabet: $\Sigma = \{1, +, =\}$

Language:

$$ADDITION = \{x + y = z : x = 1^n, y = 1^m, z = 1^k, \\ n + m = k\}$$

$$11 + 111 = 11111 \in ADDITION$$

$$111 + 111 = 111 \notin ADDITION$$

Squares

Alphabet: $\Sigma = \{1, \#\}$

Language:

$$SQUARES = \{x\#y : x = 1^n, y = 1^m, m = n^2\}$$

$$11\#1111 \in SQUARES$$

$$111\#1111 \notin SQUARES$$

Note that:

Sets

$$\emptyset = \{\} \neq \{\lambda\}$$

Set size

$$|\{\}| = |\emptyset| = 0$$

Set size

$$|\{\lambda\}| = 1$$

String length

$$|\lambda| = 0$$

Operations on Languages

The usual set operations

$$\{a, ab, aaaa\} \cup \{bb, ab\} = \{a, ab, bb, aaaa\}$$

$$\{a, ab, aaaa\} \cap \{bb, ab\} = \{ab\}$$

$$\{a, ab, aaaa\} - \{bb, ab\} = \{a, aaaa\}$$

Complement: $\bar{L} = \Sigma^* - L$

$$\overline{\{a, ba\}} = \{\lambda, b, aa, ab, bb, aaaa, \dots\}$$

Reverse

Definition: $L^R = \{w^R : w \in L\}$

Examples: $\{ab, aab, baba\}^R = \{ba, baa, abab\}$

$$L = \{a^n b^n : n \geq 0\}$$

$$L^R = \{b^n a^n : n \geq 0\}$$

Concatenation

Definition: $L_1L_2 = \{xy : x \in L_1, y \in L_2\}$

Example: $\{a, ab, ba\}\{b, aa\}$

$$= \{ab, aaa, abb, abaa, bab, baaa\}$$

Another Operation

Definition: $L^n = \underbrace{LL \cdots L}_n$

$$\{a,b\}^3 = \{a,b\}\{a,b\}\{a,b\} = \\ \{aaa, aab, aba, abb, baa, bab, bba, bbb\}$$

Special case: $L^0 = \{\lambda\}$

$$\{a, bba, aaa\}^0 = \{\lambda\}$$

$$L = \{a^n b^n : n \geq 0\}$$

$$L^2 = \{a^n b^n a^m b^m : n, m \geq 0\}$$

$$aabbbaaabb \in L^2$$

Star-Closure (Kleene *)

All strings that can be constructed from L

Definition: $L^* = L^0 \cup L^1 \cup L^2 \dots$

Example:

$$\{a, bb\}^* = \left\{ \begin{array}{l} \lambda, \\ a, bb, \\ aa, abb, bba, bbbb, \\ aaa, aabb, abba, abbbb, \dots \end{array} \right\}$$

Positive Closure

Definition: $L^+ = L^1 \cup L^2 \cup \dots$

If $\lambda \in L$ then $L^+ = L^*$

If $\lambda \notin L$ then $L^+ = L^* - \{\lambda\}$

$$\{a, bb\}^+ = \left\{ \begin{array}{l} a, bb, \\ aa, abb, bba, bbbb, \\ aaa, aabb, abba, abbbb, \dots \end{array} \right\}$$